Minho Kim

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Academic Appointments

Postdoctoral Scholar, Stanford University **Aug 2026 – Present**
Stanford Urban Resilience Initiative, Civil and Environmental Engineering
Faculty Sponsor: Jack Baker

Graduate Student Researcher, University of California, Berkeley **Jan 2022 – May 2026**
Center for Catastrophic Risk Management, Institute of Governmental Studies
River Lab, Landscape Architecture and Environmental Planning
HuMNet Lab, Civil and Environmental Engineering / City and Regional Planning

Graduate Researcher, Seoul National University **Mar 2019 – Aug 2021**
SPINS-RS Lab, Civil and Environmental Engineering
Researcher, Institute of Construction & Environmental Engineering
PR Manager, Education & Research Program (InfraSPHERE)
Lab Manager, SPINS-RS Lab

Education

University of California, Berkeley **Sep 2021 – May 2026**
Ph.D. Environmental Planning
Dissertation: Data-Driven Planning for Natural Hazard Risk Management
Advisors: Marta C. González, John Radke
Exam Committee: John Radke, Marta C. González, Iryna Dronova, Solomon Hsiang

Seoul National University **Mar 2017 – Feb 2021**
M.S. Civil & Environmental Engineering
Thesis: Local Climate Zone Classification Using Multi-Scale Convolutional Networks
Advisor: Yongil Kim

Seoul National University **Sep 2012 – Feb 2017**
B.S. Civil & Environmental Engineering
Thesis: Monitoring North Korea's 4th Nuclear Test Site with Sentinel-1A Data Using DInSAR
Advisor: Yongil Kim

Honors & Awards

Outstanding Graduate Student Instructor Award **2024**
UC Berkeley

Master's Thesis Award **2021**
Institution of Civil Engineers (UK) & Korean Society of Civil Engineers

Best Student Paper Award **2021**
Korean Society of Remote Sensing and Gaia3D at ISRS 2021

Excellence Award, Environmental Geospatial Data Idea Contest **2020**
Ministry of Environment, South Korea

Outstanding Research Award **2020**
SPINS-RS Lab, Seoul National University

Research Award, Student Competition using Meteorological Satellites **2019**
Korean Meteorological Administration

Scholarships

AEP Student Scholarship **2025**
Association of Environmental Professionals (AEP)

Beatrix C. Farrand Memorial Fellowship for Research **2024, 2025**

UC Berkeley, Dept. of Landscape Architecture & Environmental Planning
Beatrix C. Farrand Memorial Fellowship for Conference Travel **2023, 2025**
UC Berkeley, Dept. of Landscape Architecture & Environmental Planning
Robert N. Colwell Memorial Fellowship **2023**
The American Society for Photogrammetry and Remote Sensing
Brain Korea 21 Plus Scholarship **2019–2021**
National Research Foundation of Korea
Merit-based Scholarship **2014–2017, 2019**
Seoul National University
National Scholarship for Science and Engineering **2013, 2014**
Korea Student Aid Foundation
SNU Global Scholarship **2012, 2013**
Seoul National University

Publications

* Equal contribution, † Corresponding author

Working Papers

[W7] Yao, X., Wu, K., Bhatt, R., Kang, Y., & **Kim, M.** Interactive Land Cover Change Detection Using Geospatial Foundation Model Embeddings.

[W6] Domènech, R., Cunill Camprubí, **Kim, M.**, A., Artés, T., & Resco de Dios, V. Consistent fire-weather structure, divergent regimes: Pyroregion analysis of wildfires in China, 2001–2025.

[W5] **Kim, M.**†, Hosseini, M., Domènech, Castellnou, M., & González, M.C. Dynamic global pyromes under extreme fire weather risk.

[W4] Larson, C., Plass, L., **Kim, M.**, González, M.C., & Comfort, L. Simulating Catastrophic Risk in Complex Adaptive Systems. To be submitted to *Natural Hazards Review*.

[W3] **Kim, M.**, Blagojevic, N. & Baker, J. Reconstructing post-disaster recovery using multi-channel building permits. To be submitted to *International Journal of Disaster Risk Reduction*.

[W2] **Kim, M.**† Inter-visibility of wildfire surveillance camera network in California. To be submitted to *Computers, Environment, and Urban Systems*.

[W1] **Kim, M.**†, Castellnou, M., & González, M.C. Unprecedented extreme fire behavior in South Korea. Under review in *Environmental Research Letters*

Peer Reviewed Journal Papers

[P8] **Kim, M.**†, Raine, H., Radke, J., & González, M.C. A Network Approach to Mapping Responsibility in Wildfire Risk Mitigation. Accepted in *Landscape and Urban Planning*.

[P7] **Kim, M.**, Castellnou, M., & González, M.C. (2025). Modeling potential fire spread polygons and networks for suppression strategies. *International Journal of Disaster Risk Reduction*, 130, 105853

[P6] **Kim, M.**, Pais, C., & González, M.C. (2025). Fire Spread Simulations Using Cell2Fire on Synthetic and Real Landscapes. *Scientific Reports*, 15, 25173 (2025).

[P5] Yao, X., **Kim, M.**, Dronova, I., McBride, J., Kondolf, G.M., & Radke, J. (2025). Community-scale microclimate simulation using Airborne Laser Scanning and object-based urban tree classification. *Landscape and Urban Planning*, 263, 105420.

[P4] **Kim, M.**, Jeong, D., & Kim, Y. (2021). Local climate zone classification using a multi-scale, multi-level attention network. *ISPRS Journal of Photogrammetry and Remote Sensing*, 181, 345-366.

[P3] **Kim, M.**, Song, H., & Kim, Y. (2020). Direct short-term forecast of photovoltaic power through a comparative study between COMS and Himawari-8 meteorological satellite images in a deep neural network. *Remote Sensing*, 12(15), 2357.

[P2] **Kim, M.**, Jung, M., & Kim, Y. (2019). Histogram matching of Sentinel-2 spectral information to enhance Planetscope imagery for effective wildfire damage assessment. *Korean Journal of Remote Sensing*, 35(4), 517-534.

[P1] Kim, Y., **Kim, M.**, Choi, J., & Kim, Y. (2017). Image fusion of spectrally nonoverlapping imagery using SPCA and MTF-based filters. *IEEE Geoscience and Remote Sensing Letters*, 14(12), 2295-2299.

Conference & Workshop Papers and Presentations

[C16] **Kim, M.**, & González, M.C. (2026). Coupling Cell2Fire with Downscaled High Resolution Wind Data. *NHERI Computational Symposium. Berkeley, USA. May 28-29, 2026.*

[C15] **Kim, M.**, Raine, H., Radke, J., & González, M.C. (2025). A Network Modeling Approach to Quantify Homeowner Responsibility for Wildfire Risk Mitigation in the Wildland Urban Interface. *AGU 2025. New Orleans, USA. Dec 15-19, 2025.*

[C14] **Kim, M.**, Dodd, A., Radke, J., Ulyashin, V., Serra-Llobet, A., Falzone, A., & Kondolf, G.M. (2025). Flood after Fires: A Scalable Decision-Support Framework for Post-Fire Debris Flow Risk. *AGU 2025. New Orleans, USA. Dec 15-19, 2025.*

[C13] **Kim, M.**, Pais, C., & González, M.C. (2025). Fire spread simulations using Cell2Fire on synthetic and real landscapes. *International Fire Ecology and Management Congress. Association for Fire Ecology. New Orleans, USA. Dec 2-6, 2025.*

[C12] **Kim, M.**, Castellnou, M., & González, M.C. (2025). Modeling fire potential networks for suppression strategies. *Conference of Complexity Sciences 2025 (Complex Systems, Data Analytics, and Human Behavior). Siena, Italy. Sep 2-5, 2025.*

[C11] Yao, X., & **Kim, M.** (2023). A Lidar-based Method for 3D Urban Forest Evaluation and Microclimate Assessment, a Case Study in Portland, Oregon, USA. *AGU23. American Geophysical Union. Dec 11-15, 2023.*

[C10] **Kim, M.**, Dronova, I., & Radke, J. (2023). Semantic Segmentation of Enhanced Landform Maps Using High Resolution Satellite Images. *IGARSS 2023 IEEE International Geoscience and Remote Sensing Symposium. IEEE. Pasadena, California, US., July 16-21, 2023.* (*__Session Chair__)

[C9] Yao, X., & **Kim, M.** (2023). Exploratory remote sensing data analysis and clustering of urban vegetation and land surface temperature in Portland, Oregon. *IGARSS 2023 IEEE International Geoscience and Remote Sensing Symposium. IEEE. Pasadena, California, US., July 16-21, 2023.*

[C8] **Kim, M.**, Kwak, T., Jung, J., & Kim, Y. (2021). Mapping inaccessible areas using deep learning based semantic segmentation of VHR satellite images with OpenStreetMap data. *In Proceedings of the 2021 International Symposium of Remote Sensing, Virtual, May 26-28, 2021.* (*Awarded **Best Student Paper**)

[C7] **Kim, M.**, Jeong, D., Choi, H., & Kim, Y. (2020). Developing High Quality Training Samples for Deep Learning Based Local Climate Zone Classification in Korea. *arXiv preprint, Presented at **AI for Earth Sciences Workshop at NeurIPS 2020**, Virtual, arXiv:2011.01436.*

[C6] **Kim, M.**, Cho, K., Kim, H., & Kim, Y. (2020). Fusion of High Resolution Land Surface Temperature Using Thermal Sharpened Images from Regression-based Urban Indices. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 3, (pp247-254).*

[C5] Song, A., Kim, C., **Kim, M.**, & Kim, Y. (2019). Analysis of Geospatial Technology for Smart City Development: Case Study of South Korea. *In Proceedings of The 1st Tunisian Smart Cities Symposium, Tunisia, 2019.*

[C4] Kim, G., Song, H., **Kim, M.**, & Kim, Y. (2019). Multimodal Merging of Satellite Imagery with Meteorological and Power Plant Data in Deep Convolutional Neural Network for Short-Term Solar Energy Prediction. *In Proceedings of the 40th Asian Conference on Remote Sensing, Daejeon, South Korea, Oct 14-18, 2019.*

[C3] **Kim, M.**, & Kim, Y. (2019). Integration of Sentinel-2 Spectral Information with High Spatial Resolution Planetscope Imagery for Wildfire Damage Assessment. *In Proceedings of the 40th Asian Conference on Remote Sensing, Daejeon, South Korea, Oct 14-18, 2019.*

[C2] Song, H., Kim, G., **Kim, M.**, & Kim, Y. (2019). Short-Term Forecasting of Photovoltaic Power Integrating Multi-Temporal Meteorological Satellite Imagery in Deep Neural Network. *In 2019 IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC), Macao, (pp1-5).*

[C1] **Kim, M.**, & Kim, Y. (2019). Monitoring the Catastrophic 2018 Mendocino Complex Wildfire Using the Sentinel Constellation. *In Proceedings of the 2019 International Symposium of Remote Sensing, Taiwan, April 14-17, 2019.*

Other Publications (Technical Reports, White Papers, Preprints)

[O3] Ansell, C., Serra-Llobet, A., Athanasopoulos-Zekkos, A., Comfort, L., González, M.C., Keller, A., **Kim, M.**, Larson, C., Paredes, G., & Plass, L. (2025). White paper: Catastrophic Risk and System-Scale Governance. *University of California Center for Catastrophic Risk Management. Report prepared for the National Science Foundation*

[O2] Kondolf, G.M., Radke, J., Dodd, A., **Kim, M.**, Serra-Llobet, A., Falzone, A., & Ulyashin, V. (2024). Regional Sediment Bulking Methods for California in Support of Post Wildfire Flood Mitigation. *California Department of Transportation, Report No. CA24-3911*

[O1] Pais, C., **Kim, M.**, Xu, Y., Radke, J., & González, M.C. An interdisciplinary data-science approach to managing natural hazards risk. arXiv:2407.07270.

Invited Talks & Panels

[T9] **Disaster resilience and extreme wildfires**
SEMINAR *SPINS-RS Lab, Seoul National University* **Oct 2026**

[T8] **Thermal remote sensing and fire-weather interactions**
SEMINAR *Ecological Sensing Lab, Seoul National University* **Oct 2026**

[T7] **Multiscale analysis of wildfire risk in the built and natural environments**
SEMINAR *Center for Catastrophic Risk Management, UC Berkeley* **July 2026**
EXHIBITION *Just Climate Futures Exhibition, UC Berkeley* **Apr 2026**
Media: UC Berkeley | Online Exhibition

[T6] **Geospatial data science for natural hazard risk analysis**
LECTURE *CE153: Resilient Cities by Design, UC Berkeley* **April 2026**
LECTURE *457.539: Advanced Remote Sensing, Seoul National University* **Apr 2026**

[T5] **Shared responsibility and structure separation distance using network modeling and metrics**
SEMINAR *Catalan Fire Service, Forest Actions Reinforcement Group* **July 2024**
SEMINAR *Pau Costa Foundation, Spain* **July 2024**

[T4] **Fire spread modeling and superpixel-based fire suppression networks**
SEMINAR *Meteorology and Air Quality group, Wageningen University & Research* **July 2024**
SEMINAR *Catalan Fire Service, Forest Actions Reinforcement Group* **July 2024**

[T3] **Data-driven planning and modeling for wildfire research using geospatial data science and network science**
SEMINAR *Ecological Sensing Lab, Seoul National University* **Aug 2025**
LECTURE *CP4190: Introduction to Climate Change Planning, Georgia Institute of Technology* **Oct 2026**
LECTURE *CP4190: Introduction to Climate Change Planning, Georgia Institute of Technology* **Oct 2025**
LECTURE *CP4190 Introduction to Climate Change Planning, Georgia Institute of Technology* **Feb 2024**

[T2] **Exploring Research in the Environmental Field**
PANEL *Berkeley Environmental Economics and Policy Students, UC Berkeley* **Oct 2021**

[T1] **Urban Remote Sensing**
LECTURE *457.544: Satellite Image Interpretation, Seoul National University* **Apr 2020**
SEMINAR *Interdisciplinary Program in Landscape Architecture, Seoul National University* **Jan 2020**

Patents & Software

Song, H., Kim, Y., **Kim, M.**, Kim, K. Convolutional neural networks for short-term photovoltaic forecast using satellite imagery, meteorological data, and power station data. South Korea, 2021.

Teaching

Stanford University

Lead Instructor
• CEE 308: Topics in Disaster Resilience Research (*Lead Instructor*) **Fall 2026**

UC Berkeley
Lead Instructor (Course Link)
• GEOG/LDARCH C188: Geographic Information Systems (*Lead Instructor*) **Fall 2022**
- Teaching Effectiveness: 6.311/7 from 61/169 students (Department Average: 6.230/7)

Graduate Student Instructor
• LDARCH/ESPM C289: Applied Remote Sensing **Spring 2024**
• GEOG/LDARCH C188: Geographic Information Systems **Fall 2021**

Seoul National University
Teaching Assistant
• 457.542*: Advanced Surveying (*Head TA*) **Spring 2021**
• 457.205: Introduction to Geospatial Engineering (*Lab Tutor & Head TA*) **Spring 2021**
• 457.539*: Advanced Remote Sensing: VHR Imagery (*Head TA*) **Fall 2020**
• 457.402: Remote Sensing (*Lab Tutor & Head TA*) **Fall 2020**
• 457.544*: Satellite Image Interpretation (*Head TA*) **Spring 2020**
• Leadership for Civil Engineers (*TA*) **Spring 2020**
• 457.205: Spatial Informatics and Systems (*Lab Tutor & Head TA*) **Spring 2020**

Graduate-level Courses

Mentored Students at Stanford University
• Gabriela Calana Somoza (PhD Civil and Environmental Engineering) **Aug 2026 – Present**

Mentored Students at UC Berkeley
• Stella Wing (BS Conservation and Resource Studies & Data Science) **Sep 2023 – May 2024**
Current: MESM @ UCSB
• Harrison Raine (MLA & MCP) **Sep 2023 – July 2024**
Current: NASA
• Zeff Fengze Lin (Visiting student from South China University of Technology) **Jan 2023 – May 2023**
Current: Ph.D. @ Tsinghua U
• Weixin Li (MS Civil & Environmental Engineering) **Sep 2022 – May 2023**
Current: Ph.D. @ UCSB
• Xihan Yao (MLA) **Sep 2022 – May 2023**
Current: Ph.D. @ UT Austin
• Madison Chi (BS Environmental Science & Sustainable Design) **Sep 2022 – May 2023**
Current: MPH @ UCLA

Mentored Students at Seoul National University
• Hyoungwoo Choi (BS Civil and Environmental Engineering) **Sep 2020 – Feb 2021**

Services

Professional Service

Journal Reviewer (49 reviews)
Agronomy (3), Applied Sciences (2), European Journal of Remote Sensing (1), Fire (2), Forecasting (3), Geospatial Information Science (2), Geocarto International (1), GIScience & Remote Sensing (3), IEEE JSTARS (2), International Journal of Digital Earth (2), ISPRS International Journal of Geo-Information (2), Land (2), Remote Sensing (22), Sustainability (2)

Society Memberships
American Association of Geographers (2025), Association for Fire Ecology (2025), American Geophysical Union (2025), International Association of Wildland Fire (2025), Association of Environmental Professionals (2024), IEEE Geoscience and Remote Sensing Society (2023), American Society for Photogrammetry and Remote Sensing (2022), Korean-American Scientists and Engineers Association (2022), Korean Graduate Student As-

sociation (2021), International Society for Photogrammetry and Remote Sensing Student Consortium (2020), Korean Society of Civil Engineers (2019)

Student Committee, Stanford Urban Resilience Initiative **2026**

Student Committee, California Fire Science Consortium **2025, 2026**

Session Chair (Image Analysis for Land Cover Mapping), IGARSS 2023 **July 2023**

Additional Experience

- **Geospatial Data Support**, MIT Technology Review **Mar 2026 – Aug 2026**
  Advised journalists on GIS analysis and tools
- **Visiting Researcher**, Catalan Fire Service, Spain **July 2024 – Sep 2024**
  Modeled fire potential polygons and networks using fire spread simulations and hydrology-inspired tools for operational decision support; visited multiple wildfires and tested models in real-time
- **Geospatial Data Support**, Investigative Reporting Program, UC Berkeley **Sep 2022 – Dec 2022**
  Mapped deforestation and human activity in Brazil via satellite, nighttime-light, and vector data in Google Earth Engine and QGIS for the J298 OSINT Seminar; advised journalists on GIS tools
- **Ammunition Inspector**, Republic of Korea Army **May 2017 – Jan 2019**
  Logged ammunition transactions and inventory reports; contributed after-hours to a pan-sharpening image fusion study [P1]
- **Undergraduate Research Intern**, SPINS-RS Lab, Seoul National University **Aug 2016 – Feb 2017**
  Analyzed ground deformation using dInSAR with Sentinel-1 SAR imagery and monitored crop health with a ground-based hyperspectral imager
- **General Education Peer Tutor**, Seoul National University **Mar 2016 – June 2016**
  Tutored incoming freshmen in college-level English
- **Section Editor**, The SNU Quill, Seoul National University **Sep 2013 – June 2015**
  Reported and edited the campus news section across 9 volumes, overseeing 6–8 reporters
- **Research Assistant**, Lawson Health Research Institute, Canada **Sep 2011 – Jan 2012**
  Reconstructed photoacoustic images of a line source using multiple regularization percentages and maximum intensity projection in MATLAB

Projects

Roles: GSR/RA | Grant Writing | Project Manager

Awarded Projects

1. Multi-scale mitigation of wildfire risk vulnerabilities in the natural and built environments, $35,000 **2025-2026**
   College of Environmental Design, UC Berkeley
2. Deep Learning Framework for Mapping Basic Spatial Data in North Korea Using Very High Resolution Satellite Imagery, $28,000 **2021-2022**
   Institute for Peace and Unification, South Korea
3. Forecasting Photovoltaic Power using Meteorological Satellite Imagery and Deep Learning, $60,000 **2019**
   SK Telecom, South Korea

Skills

GitHub	https://github.com/minhokim93
Coding	Python, C/C++, CMake, Matlab, Javascript (Google Earth Engine), Git, Scripting (Bash), LaTeX, HTML
ML/DL	Tensorflow, Keras, PyTorch, Scikit-learn, Scikit-image, OpenCV

Remote Sensing	ENVI (SARscape), Google Earth Engine (Javascript/geemap), Python (Rasterio/GDAL)
GIS	ArcGIS, QGIS, Python
Languages	English (Native), Korean (Native), French (Fluent)

Certificates

- Teaching and Learning in Higher Education **2026**
- Geospatial Information Science and Technology **2022**

References

Dr. Prof. Jack Baker
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Dr. Prof. Marta C. González
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Dr. Prof. John Radke
Department of Landscape Architecture & Environmental Planning
Department of City and Regional Planning
University of California, Berkeley
412 Wurster Hall #2000
Email: ratt@berkeley.edu

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Lead of Forest Actions Reinforcement Group, Incident Commander
Catalan Fire Service
Avinguda de Serra Galliners, 112, 08290, Cerdanyola del Vallès, Spain
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